

Siwen Zhang

ssswzh.github.io | [LinkedIn](#) | [GitHub](#)

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CAREER OBJECTIVE

Second-year MSc Computer Science student at Leiden University, specializing in Data Science. Seeking an AI/ML internship where I can apply my background in machine learning, LLMs, and 7+ years of scientific data engineering experience to real-world AI systems and data-driven applications.

EDUCATION

Leiden University, Leiden, The Netherlands	<i>Master of Science in Computer Science</i>	Sept 2025 – Aug 2027
The University of Edinburgh, Edinburgh, United Kingdom	<i>Master of Science in Bioinformatics</i>	Sept 2015 – Aug 2016

RELATED COURSES

Agentic Large Language Models | Advances in Data Mining | Automatic Machine Learning | Evolutionary Algorithms | Introduction to Deep Learning | Reinforcement Learning | Text Mining | Information Retrieval | Recommender Systems
First year transcript in Leiden University: average 8.5/10 across 12 MSc courses

TECHNICAL SKILLS

Languages and Dev Tools	: Python, R, Shell, C (basic), SQL (basic), Docker, Git
Machine Learning	: TensorFlow, PyTorch, Keras, Sci-kit Learn, HuggingFace, LLMs
Visualization	: Matplotlib, Seaborn, ggplot2, Plotly, Tableau (basic)

PROJECTS

Robot Cat	<i>Python, ROS2, YOLO, Vosk</i>	Source Code
Developed a ROS 2-based interactive robotic cat integrating YOLO vision, speech commands, finite-state behavior and expressive actuator control. Implemented person following, object search, docking, and state-based behaviors.		
Reinforcement Learning Algorithms for CartPole Problem	<i>Python, TensorFlow, Keras</i>	
Implemented and compared Naive DQN, REINFORCE, AC, A2C, and PPO in CartPole environment. Several engineering tricks (mini-batch, GAE, etc.) were applied to improve performance and stability.		
Recommendation Challenge for Amazon Products	<i>Python, Kaggle, PyTorch</i>	
Built an Amazon Top-10 product recommendation system using SASRec-based sequential modeling and metadata-aware candidate filtering, achieving 8th/60 place on the Kaggle leaderboard.		
Black-box Problem Optimization using Evolution Algorithm	<i>Python</i>	
Implemented and evaluated Genetic Algorithms and self-adaptive Evolution Strategies for discrete and continuous black-box optimization, with budget-aware hyperparameter tuning and convergence analysis.		

EXPERIENCE SNAPSHOT

Middle Bioinformatics Engineer	<i>Feb 2017 – Sept 2024</i>	Beijing, China
- Delivered MRD/HRD analytics for oncology studies, built platelet RNA-seq ML classifiers, supported 5 cell and gene therapy products, and productionized cloud workflows that reduced runtime and cost by 50%. - Led the first long-read targeted sequencing development for rare disease diagnosis in China, built COVID-19 pathogen surveillance workflows, and benchmarked Nanopore, PacBio, and Illumina platforms. - Developed oncology biomarker assays and reproducible genomic workflows for CNV, MSI, TMB, survival analysis, and multi-omics interpretation.		

CERTIFICATIONS

- [IBM Data Science Professional Certificate](#)
- [5 published articles and 3 patents](#)